

Weekly Progress Report

Amity School of Engineering & Technology

Project Group ID - 265

**Project Area – Data Science
(Machine Learning)**

B. Tech (8CSE-DS-1Y) VII Semester

WPR No. 1

Duration: 08/12/2025 to 14/12/2025

Students Weekly Progress Report (WPR) for Even Semester of session 2025-2026

To be filled by Students	
Students Name	1. Sagar Singh Bisht 2. Aakarsh Pab
Roll no.	1. 056 2. 050
Enrollment no.	1. A023167022056 2. A023167022050
Project Title finalized, if Yes, give name, if NO, give reason	Machine Learning based Predictive Model for Livability using Environmental and Socioeconomic indicators
Synopsis submitted	YES
Literature review	YES
Technical & Economical Feasibility	YES
Bill of Material	NA
Project Progress Schedule (PERT Chart)	NA
Design of critical components	-
Fabrication work (give %)	-
Experimental work (give %)	-
Result and Analysis	-
Report writing	-
Signature of students	

Work done in this week:

Literature Survey & Problem Pivoting

1. Problem Redefinition (Responding to Examiner Feedback)

- **Critique Analysis:** Analyzed the feedback regarding the previous "Predictive Model." The core issue was redundancy (users can already check AQI on phones) and the arbitrary nature of the "Livability Score" (0-100).
- **Strategic Pivot:** Shifted the project objective from a **Descriptive** approach (telling users *current* conditions) to a **Prescriptive** approach (recommending *where* to live).
- **New Paradigm:** Defined the "CityMatch" concept. Instead of a "one-size-fits-all" score, the system will now treat urban relocation as a **Multi-Objective Optimization Problem**, where "livability" is defined dynamically by the user's specific constraints (e.g., maximizing air quality vs. minimizing rent).

2. Literature Survey: Technical Framework

Conducted a targeted review of Machine Learning algorithms suitable for "Unsupervised" pattern discovery and recommendation.

- **A. Unsupervised Learning (Clustering)**
 - **Algorithm Selected: K-Means Clustering.**
 - **Rationale:** Unlike Supervised Learning (which requires labeled "correct" answers), K-Means allows the system to discover hidden structures in the data naturally.
 - **Application:** We will use this to group Delhi's neighborhoods into distinct "Archetypes" (e.g., *Cluster 1: High Affordability & Industrial*, *Cluster 2: Premium Infrastructure & Green Spaces*). This creates a "map" of the city's structure without human bias.
- **B. Recommender Systems (Content-Based Filtering)**
 - **Algorithm Selected: Cosine Similarity (Vector Space Model).**
 - **Rationale:** We rejected "Collaborative Filtering" (which requires user history data like Netflix) in favor of "Content-Based Filtering" (which matches item attributes).
 - **Application:**
 - **User Vector (A):** Represents user priorities (e.g., [Air: 0.9, Rent: 0.2, Metro: 0.8]).
 - **Neighborhood Vector (B):** Represents actual data (e.g., [Air: 0.85, Rent: 0.4, Metro: 1.0]).

- **The Match:** The system calculates the cosine of the angle between vectors A and B. A smaller angle implies a higher "Compatibility Score."

3. Feasibility Study: Data Acquisition

- **Gap Identification:** The previous dataset (Air Quality + Traffic) was insufficient for clustering.
- **Solution Identified: OSMnx** (Python Library for OpenStreetMap).
- **Capabilities Confirmed:** Research confirmed that OSMnx can programmatically download street networks and "Points of Interest" (POIs). This allows us to generate new features previously missing, such as:
 - *Education Density:* Count of schools within 1km.
 - *Health Accessibility:* Count of hospitals/clinics.
 - *Green Cover:* Area of parks/gardens.

4. Conclusion & Next Steps

The theoretical foundation for our Livability model is established. The project has successfully pivoted from a basic regression task to a sophisticated recommendation system.

- **Next Immediate Goal:** Set up the Data Acquisition Pipeline (Task 2) to extract the amenity data using OSMnx.

To be filled with a Guide (strike off whichever is not applicable)	
The performance of students is satisfactory	
The performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 12/08/2025	Signature of Guide